

Advanced techniques in applied economics

Lecture 6: Unobserved confounding and adjustments

Piotr Zwiernik



**Universitat
Pompeu Fabra**
Barcelona

Statistics, Probability
and Machine Learning
Research Group



Barcelona School of Economics

Spring 2026

When the clean DAG picture breaks

What changes today?

In the previous lectures we mostly acted as if the important variables were observed. Today we ask what happens when a relevant common cause is hidden.

Main difficulty

If an important confounder is unobserved, then

- back-door adjustment may fail,
- graphs on observed variables may be misleading,
- standard regression can be badly biased.

Main response

We cannot make hidden confounding disappear, but we can often reason about it and partially adjust for it:

- instrumental variables,
- proxy controls,
- latent factor adjustments,
- synthetic instruments.

Main question: How can we reason causally when the structural graph contains hidden variables?

Roadmap for today

1. hidden confounding and why the clean DAG story breaks
2. a brief glimpse of MAGs
3. IV as the benchmark workaround
4. proxy controls
5. latent factor adjustments
6. synthetic-instrument style ideas

Memorable statement: when a confounder is hidden, the graph is not gone — it is only partially visible.

Part 1: Hidden confounding

The clean adjustment story

Suppose we want the effect of treatment D on outcome Y .



Adjustment logic

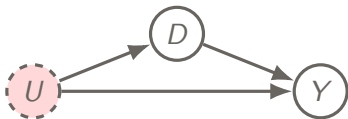
If X blocks all back-door paths from D to Y , then adjusting for X may identify the causal effect.

This is the ideal case

The confounder is observed, and the graph tells us what to condition on.

What if the confounder is hidden?

Now suppose the true common cause U is unobserved.



Main problem

The back-door path

$$D \leftarrow U \rightarrow Y$$

is open, but we cannot condition on U because it is hidden.

Economic reading

U could be ability, firm quality, local demand, risk preferences, health, or political pressure.

A regression view of omitted-variable bias

Suppose

$$Y = \alpha D + h(W) + \lambda U + \varepsilon, \quad D = s(W) + \rho U + \nu,$$

where U is unobserved.

After partialling out W , the residualized equations are

$$\tilde{Y} = \alpha \tilde{D} + \lambda \tilde{U} + \varepsilon, \quad \tilde{D} = \rho \tilde{U} + \nu.$$

What goes wrong?

The remaining variation in D is correlated with the omitted term:

$$\text{Cov}(\tilde{D}, \lambda \tilde{U} + \varepsilon) = \lambda \rho \text{Var}(\tilde{U}) \neq 0.$$

Lesson

Even after controlling for observed covariates, naive regression can still be biased because hidden structure remains in the error.

A tiny omitted-variable simulation

Suppose

$$D = U + \varepsilon_D, \quad Y = 1.0 D + U + \varepsilon_Y.$$

```
set.seed(1)
n <- 3000
U <- rnorm(n)
D <- U + rnorm(n)
Y <- 1.0 * D + U + rnorm(n)

coef(lm(Y ~ D))
```

```
# impossible benchmark:
# if U were observed
coef(lm(Y ~ D + U))
```

Interpretation

The naive regression on D alone mixes the true causal effect with hidden confounding from U .

Part 2: A brief glimpse of MAGs

Why introduce MAGs?

If latent variables are present, the observed variables alone may no longer form a DAG with the correct conditional independence structure.

What a MAG does

A **maximal ancestral graph** (MAG) is a graph on the observed variables that summarizes

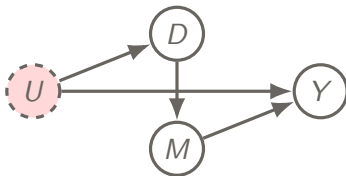
- directed relations that remain visible,
- dependence induced by hidden common causes,
- which variables can still be separated by conditioning.

For this course

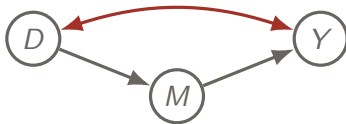
We only need the intuition: after marginalizing over hidden variables, the correct observed graph is often a mixed graph rather than a clean DAG.

From a hidden-variable DAG to a mixed graph

Suppose the hidden DAG is



After hiding U , we may summarize the observed structure by



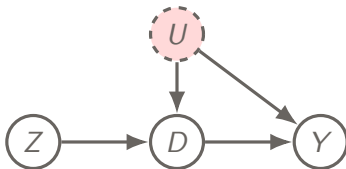
Reading the graph

The bidirected edge $D \leftrightarrow Y$ is a reminder that the two observed variables still share hidden confounding.

Part 3: IV as the benchmark workaround

IV intuition

If the confounder is hidden, the classical econometric workaround is to find an observed variable Z that moves the treatment but is otherwise exogenous.



IV requirements

- **Relevance:** Z affects D .
- **Exogeneity:** Z is independent of the hidden confounder.
- **Exclusion:** Z affects Y only through D .

Applied econ examples of IVs

Education and earnings

Use distance to college, local college openings, or schooling reforms to move education without directly moving earnings.

Health and detention

Use physician or judge assignment when assignment is as-good-as random and strictness differs across decision-makers.

Military service

Use draft lottery numbers to generate exogenous variation in service.

Main lesson

IV is often the cleanest tool when hidden confounding is serious and a credible instrument is available.

A tiny IV simulation

Suppose

$$D = 0.8Z + U + \varepsilon_D, \quad Y = 1.0D + U + \varepsilon_Y,$$

where Z is independent of U .

```
set.seed(1)
n <- 4000
Z <- rnorm(n)
U <- rnorm(n)
D <- 0.8*Z + U + rnorm(n)
Y <- 1.0*D + U + rnorm(n)

coef(lm(Y ~ D))
```

```
library(AER)
coef(ivreg(Y ~ D | Z))
```

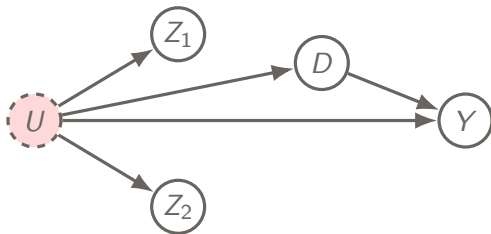
Interpretation

OLS is biased by the hidden confounder U . A valid IV can recover the causal effect using exogenous variation in D coming from Z .

Part 4: Proxy controls

The basic proxy idea

Even if we do not observe the confounder U , we may observe variables that are informative about it.



Idea

The proxies Z_1, Z_2 are not themselves the confounder, but they carry information about it.

Applied examples

Test scores, prior grades, neighborhood variables, balance-sheet variables, or baseline outcomes may all serve as noisy measurements of an unobserved confounder.

One proxy vs several proxies

A single noisy proxy is often not enough to remove confounding.

One proxy

If $Z = U + \text{noise}$,
then conditioning on Z may reduce
confounding, but not eliminate it.

Several proxies

With multiple noisy measurements of the
same latent source, we can often extract a
much better approximation to the hidden
confounder.

A good proxy does not have to be perfect. It has to carry usable information about the hidden source of confounding.

A tiny proxy-control simulation

Suppose

$$Z_1 = U + \eta_1, \quad Z_2 = U + \eta_2, \quad D = U + \varepsilon_D, \quad Y = 1.0 D + U + \varepsilon_Y.$$

```
set.seed(1)
n <- 3000
U <- rnorm(n)
Z1 <- U + rnorm(n)
Z2 <- U + rnorm(n)
D <- U + rnorm(n)
Y <- 1.0*D + U + rnorm(n)

coef(lm(Y ~ D))
```

```
coef(lm(Y ~ D + Z1))
coef(lm(Y ~ D + Z1 + Z2))
```

Interpretation

One proxy helps a bit; multiple proxies help more. The point is not perfection, but partial recovery of the hidden confounder.

Part 5: Latent factor adjustments

Latent factor adjustments

If the hidden confounding is approximately low-dimensional, we may try to estimate it through a factor model.

Simple idea

Suppose many observed variables are affected by the same hidden source U . Then a factor model may recover a low-dimensional summary of that hidden variation.

Adjustment strategy

Estimate the latent factor(s), then include the estimated factors as additional controls in the outcome model.

Applied reading

This is natural when hidden confounding comes from business-cycle conditions, latent productivity, local demand, or market-wide stress.

A tiny latent-factor adjustment in R

```
set.seed(1)
n <- 2500
U <- rnorm(n)

X1 <- 0.8*U + rnorm(n)
X2 <- 0.7*U + rnorm(n)
X3 <- 0.9*U + rnorm(n)
D <- U + rnorm(n)
Y <- 1.0*D + U + rnorm(n)

X <- cbind(X1, X2, X3)
```

```
pc1 <- prcomp(X, scale. = TRUE)$x[,1]

coef(lm(Y ~ D))
coef(lm(Y ~ D + pc1))
```

Interpretation

The first principal component acts as a rough estimate of the hidden confounder and can reduce omitted-variable bias.

Part 6: Synthetic instruments and practical strategies

Synthetic instruments: the basic idea

Sometimes there is no obvious external instrument. But we may have many related variables that are informative about the treatment mechanism.

Heuristic idea

Construct a derived variable from multiple observed measurements so that it is:

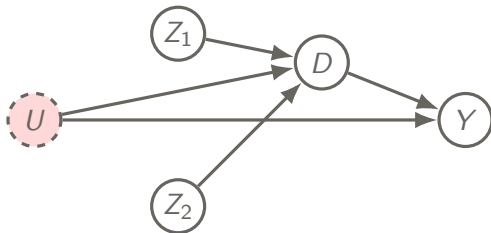
- strongly related to the treatment,
- but plausibly less contaminated by the hidden confounder than the raw treatment itself.

Why the name?

The constructed variable is not a classical external instrument. It is a synthetic or model-based source of variation extracted from the data.

A stylized synthetic-instrument picture

Suppose a hidden factor U affects treatment and outcome, but we also observe many variables tied to the treatment mechanism.



Idea

A suitable combination of $Z_1, Z_2, \dots$ may isolate a more credible source of variation for D than the raw treatment itself.

A pragmatic summary

When the graph contains hidden variables, there is no universal fix.

Good practical strategies

- use substantive knowledge,
- search for multiple proxies,
- exploit repeated measurements,
- use factor structure when confounding is common and pervasive,
- use IV if a credible instrument exists.

Things to avoid

- pretending the hidden variable is irrelevant,
- conditioning on obvious colliders,
- over-interpreting a graph estimated from incomplete variables,
- assuming one noisy proxy solves the whole problem.

The right question is often not how do I remove confounding completely? but how much credible adjustment can I recover from the observed data?

Take-away

- Hidden confounders break the clean adjustment logic of observed-variable DAGs.
- MAGs give a compact language for the observed implications of latent confounding.
- IV is the benchmark workaround when a credible instrument is available.
- Proxy controls use noisy measurements of the hidden confounder.
- Latent factor adjustments use many observed variables to estimate a hidden source.
- Synthetic-instrument ideas use structured observed variation to recover more credible identification.

When the graph is only partially observed, the goal shifts from perfect recovery to careful partial adjustment.

Looking ahead

Next lecture

We move from hidden confounding to interactions between units: network interference and spillovers.

Bridge forward

Today hidden variables complicated identification within a unit. Next time other units will enter directly into the potential outcome.